

Compact Near-Field Metasurface for 2-D Antenna Beam Tilting

Bo Zhang[✉], Member, IEEE, and Zhi Ning Chen[✉], Fellow, IEEE

Abstract—A compact near-field metasurface (NFM) for 2-D beam tilting of a single antenna is crucial and challenging, as the design involves complex near-field electromagnetic behavior. This article proposes a theoretically grounded method for designing NFMs for antenna beam tilting. A thorough investigation into the contribution of near-field components of a dipole antenna to far-field radiation reveals that the parallel-polarized components exhibit an approximately fixed phase difference between the near fields and far fields of the Hertzian dipole. The study demonstrates the feasibility of tailoring the phase gradient extracted from near fields to achieve beam tilting. Building on this theoretical understanding, phase-gradient NFMs composed of ideal materials are designed to cover a single antenna for beam tilting in both elevation and azimuth planes. Prephase offsets are introduced to mitigate edge diffraction and guided-wave resonances. Subsequently, a phase-gradient NFM is designed to be integrated with a half-wave dipole. A PCB-based NFM prototype with dimensions of $\lambda_0 \times \lambda_0$ is mounted $0.05\lambda_0$ right above a dipole antenna operating at 2 GHz. The results validate theoretical findings, demonstrating beam tilting at $\theta_0 = 45^\circ$ and $\varphi_0 = 90^\circ$, demonstrating the potential of compact NFMs for antenna radiation control.

Index Terms—Antenna beam tilting, metasurface (MTS), near fields.

I. INTRODUCTION

CONVENTIONAL single antennas face challenges in adapting to complex environments, while beam tilting techniques provide a flexible solution for redirecting radiation. However, achieving beam tilting for a single antenna within a compact and closely spaced aperture remains challenging.

Traditional antenna beam tilting is achieved by mechanical adjustment [1], [2], [3], where the antenna or its supporting structure is physically adjusted to redirect the beam. While effective, this approach is constrained by limited adaptability, increased structural complexity, and a higher risk of interference with nearby antennas. A well-established alternative is the use of parasitic elements, such as directors and reflectors, to influence the radiation pattern of a driven element [4], [5], [6], [7]. For example, additional reflectors are introduced near the excited antenna to reflect the electromagnetic waves and thereby control the radiated wave [4], [5]. Meanwhile, directors carefully placed near the driven antenna contribute to guiding the radiated wave and controlling the radiation pattern [6], [7]. Recently, metamaterials [8], [9], [10], [11] have been loaded near antennas to manipulate propagation constants and

wave fronts for radiation beam tilting. These designs are typically analyzed under the far-field approximation. Although effective in controlling radiation patterns, especially in a 1-D plane, the precise realization of the desired beam tilt often requires a process of trial and error.

Besides loadings, metasurfaces (MTSs) have been proposed to manipulate the beams of antennas [12], [13] since generalized Snell's law [14]. With the assumption of plane or spherical wave incidence, the MTSs are typically placed a few wavelengths away from the antennas. For compactness and low-profile, studies of near-field MTSs (NFMs) are gaining increasing attention [15], [16], [17], [18], [19], [20], [21], [22], [23]. One milestone in this area is the time-delay equalizer MTS, proposed in [15], which numerically computed nonuniform phase data from an antenna for lateral phase correcting structures. Numerous studies have been devoted to the NFM design based on this idea, and achieved advanced beam control capabilities as gain enhancement [15], [16], [18] and beam steering [17], [19] with the distance between the antenna and the MTS being less than $0.5\lambda_0$.

These works neglect the near-field components, which is a fair approximation. Since a partially reflecting surface placed $0.5\lambda_0$ above the driven antenna forms a cavity in which plane-wave components dominate. In addition, examples of antenna arrays with near-placed MTSs can be found for beam steering [20], [21], [22], where the wavefront phase is naturally more uniform due to the interaction among antenna elements. Despite these advancements, existing designs typically rely on a separation distance exceeding $0.5\lambda_0$ between the antenna and the MTS. This spacing is widely adopted to mitigate effects from reactive near fields or to form Fabry–Perot cavities. Meanwhile, implementing radiation beam control in such systems frequently necessitates electrically large apertures to receive enough incident energy, restricting their utility in compact, space-limited environments.

Modeling and analyzing the near fields of an antenna remain challenging. The near fields can be numerically calculated by electromagnetic numerical algorithms, such as FDTD, FEM, and MoM [24]. However, near-field behavior and interaction with materials are obscure. A visualization method [25], [26] has been proposed to draw Poynting streamlines near antennas, clearly revealing the energy flow and providing new physical insights. To model the near fields of antennas, Huygens sources are widely employed to form surfaces composed of electric and magnetic Hertzian dipoles [27], [28], [29], [30], [31]. Moreover, the interaction of dielectric spheres with the near fields is studied in [32] and [33]. This motivates a perspective of modeling unit cells as scatterers excited by antenna near fields.

This article investigates the near-field components of a dipole antenna to design a compact NFM for 2-D beam tilting, as illustrated in Fig. 1. Distinct from previous approaches

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The authors are with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore 117583 (e-mail: bzhang@nus.edu.sg; eleczn@nus.edu.sg).

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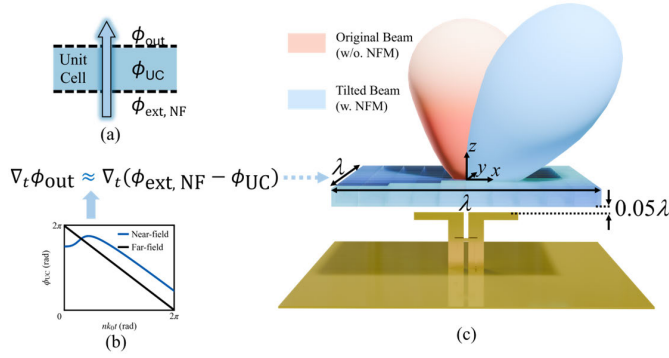


Fig. 1. Schematic of the proposed NFM for beam tilting. (a) Illustration of the E-field phases at different interfaces. The E-field phase at the incident interface of the unit cell is extracted from the simulated results of a dipole antenna, denoted as $\phi_{\text{ext, NF}}$. ϕ_{UC} is the transmitted phase of the unit cell, and ϕ_{out} is the output phase from the unit cell. (b) Phase gradient of the output phase $\nabla_t \phi_{\text{out}}$ can be approximately calculated by the extracted near-field phase $\phi_{\text{ext, NF}}$. (c) Configuration of the proposed compact NFM for beam tilting.

that rely on large separation distances or bulky structures, the proposed NFM realizes precise 2-D beam tilting with an aperture of only $\lambda_0 \times \lambda_0$, positioned at an extreme proximity of $0.05\lambda_0$ from the antenna.

The remainder of this article is organized as follows. Section II analyzes the near-field characteristics of a dipole antenna, establishing guidelines for designing compact NFMs by leveraging parallel-polarized components. Section III demonstrates the feasibility of achieving 2-D beam tilting based on extracted near-field phases, as depicted in Fig. 1(a) and (b). Section IV validates this concept using an NFM composed of ideal materials, corresponding to the schematic shown in Fig. 1(c). Finally, Section V presents the fabrication and measurement of a prototype using PCB unit cells to experimentally verify the designed beam tilting performance.

II. STUDY OF NEAR FIELDS OF A DIPOLE ANTENNA

Consider a half-wave dipole antenna placed at a height $h = \lambda_0/4$, where λ_0 is the operation wavelength, above a ground plane, its near fields are simulated and extracted on a plane positioned at a distance d above the dipole and parallel with the ground, as illustrated in Fig. 1.

For dipole antennas, electric energy predominates in the near-field region and is the focus of the following analysis. The time-averaged energy density $w_{e,i}$ of each component of the electric field (E-field) in the extracted plane can be calculated by [24]

$$w_{e,i} = \frac{1}{4} \varepsilon |E_i|^2 \quad (1)$$

where $i = x, y, \text{ or } z$. ε is the permittivity. The total energy on the extracted plane S can be integrated by

$$W_{e,i} = \iint_S w_{e,i} dS = \frac{1}{4} \iint_S \varepsilon |E_i|^2 dS. \quad (2)$$

In Fig. 2, $W_{e,x,l}$ is the electric energy within the $l \times l$ square region in the extraction plane, while $W_{e,x}$ is the electric energy in the entire extraction plane. The ratios of $W_{e,x,l}$ to $W_{e,x}$ are calculated for varying d and different l . Suppose that an NFM is placed in the $l \times l$ square region, the results in Fig. 2 demonstrate that either decreasing d or increasing l leads to a higher ratio $W_{e,x,l}/W_{e,x}$, indicating an enhanced energy

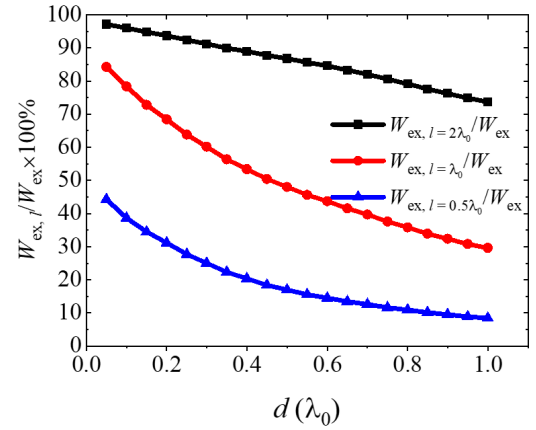


Fig. 2. Ratios from $W_{e,x,l}$ to $W_{e,x}$ of the whole extracted plane with respect to d .

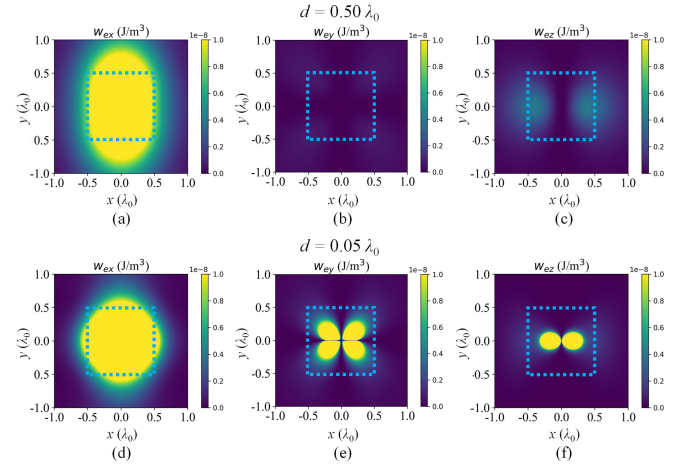


Fig. 3. Time-averaged electric energy density w_e of each component: (a) w_{ex} on $d = 0.50\lambda_0$, (b) w_{ey} on $d = 0.50\lambda_0$, (c) w_{ez} on $d = 0.50\lambda_0$, (d) w_{ex} on $d = 0.05\lambda_0$, (e) w_{ey} on $d = 0.05\lambda_0$, and (f) w_{ez} on $d = 0.05\lambda_0$. (The blue-dotted squares denote a $\lambda_0 \times \lambda_0$ area.)

interaction between the antenna and the NFM within the specified region. Notably, when $d = 0.05\lambda_0$ and $l = \lambda_0$, 84.2% of $W_{e,x}$ passes through the area. This demonstrates that NFM designs can achieve compactness while maintaining sufficient energy interaction. Consequently, $d = 0.05\lambda_0$ and $l = \lambda_0$ are adopted for subsequent analysis and design of compact NFMs.

Fig. 3 compares the distribution of w_e at $d = 0.50\lambda_0$ and $0.05\lambda_0$. The blue-dotted squares in Fig. 3 denote a $\lambda_0 \times \lambda_0$ area right above the antenna. Compared with Fig. 3(a)–(c), electric energy within the plane of $d = 0.05\lambda_0$ is more concentrated in the squares, verifying the previous conclusion and enabling compact NFM designs. In addition, given that the dipole antenna is x-polarized, the parallel-polarized component E_x dominates in the far-field region along the broadside direction, with the other two components being nearly negligible, as shown in Fig. 3(a)–(c). In contrast, on the near-field cutting plane with $d = 0.05\lambda_0$, depicted in Fig. 3(d)–(f), all three components exhibit significant distributions. Specifically, $W_{e,x}$, $W_{e,y}$, and $W_{e,z}$ account for 48.8%, 30.9%, and 20.3% of the total electric energy W_e , respectively, highlighting the necessity of examining their individual contributions to the far-field region.

For an ideal half-wavelength dipole antenna along the x -axis, the E-fields in the far-field zone are approximated by [34]

$$\mathbf{E} \approx \hat{\mathbf{u}} j\eta \frac{I_0 e^{-jkr}}{2\pi r} \left[\frac{\cos\left(\frac{\pi}{2} \cos\psi_x\right)}{\sin\psi_x} \right]$$

$$\psi_x = \arccos(\sin\theta \cos\varphi) \quad \text{and} \quad \hat{\mathbf{u}} \equiv \frac{(\hat{\mathbf{r}} \times \hat{\mathbf{a}}_x) \times \hat{\mathbf{r}}}{|(\hat{\mathbf{r}} \times \hat{\mathbf{a}}_x) \times \hat{\mathbf{r}}|} \quad (3)$$

where η is the wave impedance in free space and I_0 denotes the current amplitude of the dipole. The polar angle ψ_x is the angle between the unit radial vector $\hat{\mathbf{r}}$ and the unit vector $\hat{\mathbf{a}}_x$ along the x -axis. θ and φ denote the polar and azimuth angles, respectively, in the spherical coordinate system. The unit vector $\hat{\mathbf{u}}$ denotes the polarization direction, defined in the plane with the normal vector $\hat{\mathbf{n}} = \hat{\mathbf{r}} \times \hat{\mathbf{a}}_x$ and perpendicular to the radial unit vector $\hat{\mathbf{r}}$.

In image theory, the field distribution for a dipole placed a quarter-wavelength above an infinite ground plane can be easily derived. The time-averaged electric energy density ratios of y - and z -components to the x -component are compared across various directions, as illustrated in Fig. 4. In conventional MTS designs, beam tilting within $\theta < 45^\circ$ is of primary interest. Within this region, the analysis reveals that w_{ey} is one order of magnitude smaller than w_{ex} , whereas w_{ez} becomes more significant, particularly near $\theta = 0^\circ$ and 180° . Assuming that the constitutive tensor of the NFM includes only diagonal elements and achieves full transmission, there is no polarization conversion, and these ratios remain consistent after applying the NFM. Therefore, the contribution of the y -component in the far fields almost vanishes in the interested area.

Furthermore, the energy of the E_z component in the near-field zone of a dipole antenna is nonnegligible. However, controlling E_z requires manipulation of the normal constitutive relations of the unit cells, imposing a stringent constraint on unit cell design based on PCB technology. Given the electrically thin nature of the MTS and the constant normal relative permittivity of the unit cells (set to 6.15), a reasonable approximation is that the scattered field excited by E_z is dominated by the electric dipole mode, which does not radiate power along the z -axis but reaches the maximum in the horizontal (xy) plane. Moreover, the power of E_z component accounts for only 20.3% of the total E-field power. Therefore, the contribution of E_z -induced scattering fields to the overall radiation pattern, especially for relatively small beam tilting polar angles, is limited. Thus, its impact on the far-field radiation pattern can be considered negligible in the NFM design.

In summary, the analysis of near-field energy distribution provides the physical guideline for ensuring efficient energy interaction and radiation control.

- 1) *Spatial Energy Interception*: As quantified in Fig. 2, positioning the NFM at a proximity of $d = 0.05\lambda_0$ allows the compact aperture ($\lambda_0 \times \lambda_0$) to spatially intersect with over 84% of the total radiated energy. This high interception ratio is the fundamental mechanism for effective beam manipulation, ensuring that the MTS can interact with the majority of the source power in a compact aperture.
- 2) *Dominant Polarization Control*: Since the parallel-polarized component E_x accounts for the dominant contribution to the far-field radiation, as shown in Fig. 4,

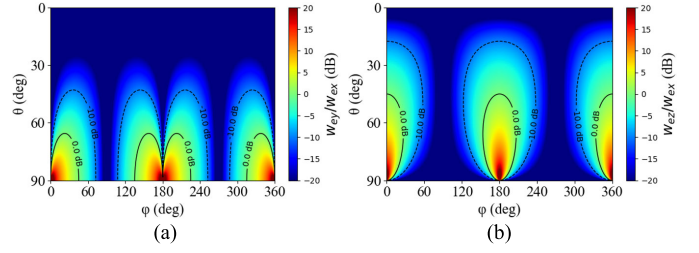


Fig. 4. Time-averaged electric energy density ratios in the far-field region of a half-wavelength dipole above an infinite ground plane as a function of observation angle: (a) w_{ey}/w_{ex} and (b) w_{ez}/w_{ex} .

the design can be simplified by focusing solely on the manipulation of the E_x component.

This establishes the physical basis for the NFM design. By capturing the concentrated near-field energy and targeting the dominant polarization, the system maximizes the potential for precise radiation control, which will be realized through the phase-gradient design discussed in Section III.

III. NFM DESIGN

As previously stated, the energy of the parallel-polarized component (E_x) dominates in the near-field region. Accordingly, the NFM design focuses on controlling the E_x component to enable beam tilting functionality. Phase gradient along an MTS is one of the most effective approaches for controlling antenna radiation beams. In conventional scenarios, the term “phase” refers to the phase of electric field components that contribute to the far fields, i.e., radiated components. In these cases, the MTS primarily imposes a desired phase shift for beam tilting. However, in the near-field region, the extracted E-field phases contain both radiated and nonradiated components simultaneously, which are difficult to distinguish in full-wave simulation results. Directly implementing a phase-gradient MTS based on the extracted near-field phase distribution lacks a clear theoretical basis for ensuring the desired phase transformation of only the radiated field components. To address this challenge, an equivalent surface is established beneath the NFM to fully enclose the dipole antenna and its ground plane. According to Love’s equivalence principle [34], any sources within the enclosed volume can be represented by equivalent electric and magnetic surface current densities, effectively acting as Hertzian dipoles, distributed over this boundary. In a homogeneous medium characterized by a phase constant k , the radiation from an electric Hertzian dipole is expressed as

$$\mathbf{E} = \hat{\mathbf{r}} \cdot \frac{\eta I_l \cos\psi_x}{2\pi r^2} \left(1 + \frac{1}{jkr}\right) e^{-jkr} + \hat{\mathbf{u}} \cdot \frac{jk\eta I_l \sin\psi_x}{4\pi r} \left(1 + \frac{1}{jkr} - \frac{1}{(kr)^2}\right) e^{-jkr}. \quad (4)$$

Note that at each observation point on the extraction plane, $\hat{\mathbf{a}}_x \cdot \mathbf{E}$ is proportional to $\hat{\mathbf{u}} \cdot \mathbf{E}$ with no phase difference. Then, the phase of $\hat{\mathbf{u}} \cdot \mathbf{E}$ is evaluated to represent the phase of E_x in this section. Unlike the far-field expression, where only the r^{-1} term remains significant. The E-field phase is solely determined by the phase term e^{-jkr} , and the E-field phase in the near-field region is additionally influenced by the complex

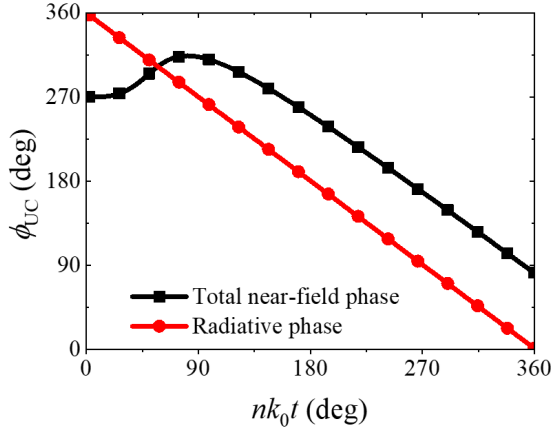


Fig. 5. Transmitted phase of the Hertzian dipole through a unit cell as a function of nk_0t : total near-field phase versus radiative-only phase.

term $[1 + 1/(jkr) - 1/((kr)^2)]$. Therefore, the transmitted near-field phase of transverse component ϕ_{UC} is thus calculated by

$$\phi_{UC} = \arctan \left(\frac{\text{Im}(\hat{\mathbf{u}} \cdot \mathbf{E})}{\text{Re}(\hat{\mathbf{u}} \cdot \mathbf{E})} \right) = \delta - kr$$

$$\delta = \arctan(kr - (kr)^{-1}) \quad (5)$$

where δ is the phase contributed by the near-field components. Suppose that a unit cell with refractive index n is placed near the source, and then, the transmission phase passing through the NFM with thickness t is

$$\phi_{UC} = \delta_t - nk_0t = \arctan(nk_0t - (nk_0t)^{-1}) - nk_0t. \quad (6)$$

The transmitted phases with and without the near-field components are calculated and compared as plotted in Figs. 1(b) and 5, where t is assumed to be $\lambda_0/15$. From (6), $\delta_t = -\pi/2$, which represents the phase difference between the total near-field and radiated field components when nk_0t is relatively small, as shown in Fig. 5. Similarly, $\delta_t = \pi/2$, indicating the $\pi/2$ phase difference when nk_0t exceeds $\pi/2$.

In the range of $\pi/2 < nk_0t \leq 2\pi$, the phase slope of the near fields closely aligns with that of the radiated field. This phase slope consistency serves as the physical basis for resolving the ambiguity between radiative and reactive field components. Since the beam tilting angle is fundamentally determined by the spatial derivative of the phase according to the generalized Snell's law, rather than the absolute phase value, the design does not require an explicit decomposition of field components. By imposing the desired phase gradient on the total near field, which is dominated by the component E_x as discussed in Section II, the NFM effectively manipulates the wavefront of the radiative component to achieve the intended beam deflection.

Overall, the extracted phase of E_x at each point on the NFM can be interpreted as the superposition of contributions from all infinitesimal current sources on the dipole. Each contribution exhibits the same phase gradient as the far-field components. Although distinguishing between reactive near-field components and radiated far-field components is challenging, the key insight is that the desired phase gradient is preserved. If the output phases of the unit cells are designed to follow the required gradient, the resulting far-field phases also exhibit the intended gradient, even though excited by the near fields. Consequently, the radiated beam is tilted as desired.

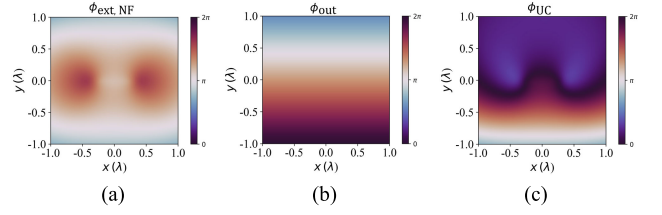


Fig. 6. (a) Extracted phase of E_x on a plane with $d = 0.05\lambda_0$. (b) Designed phase to achieve beam tilting toward $\theta = \pi/4$ and $\varphi = \pi/2$. (c) Required transmitted phase distribution of the NFM.

It is worth noting that while the derivation above utilizes a scalar refractive index n , the conclusion is equally applicable to anisotropic unit cells, which are common in practical implementations. As demonstrated in Section II, the near-field interaction is predominantly determined by the parallel-polarized component E_x . Consequently, for anisotropic media, the scalar n in (6) essentially corresponds to the effective refractive index governing the x -polarized wave component. Thus, the proposed phase-gradient design remains robust even when using unit cells with anisotropic constitutive parameters.

IV. NFM BY IDEAL MATERIAL

To achieve the desired gradient phase shift, unit cells composed of ideal materials with $\epsilon_{tr} = \mu_{tr} = n$ are used to construct an NFM, while the normal components are fixed at $\epsilon_{zr} = 6.15$ and $\mu_{zr} = 1$ to facilitate practical implementation using PCB technology. First, the E-field components of a half-wave dipole are extracted in a plane located at $d = 0.05\lambda_0$ above the dipole, as shown in Fig. 3(d) and (e). As previously discussed, E_x contributes most to the far-field radiation. Therefore, the phase of E_x is considered in this section, as shown in Fig. 6(a). To achieve beam tilting toward $\theta = \theta_0$ and $\varphi = \varphi_0$, the output phase distribution ϕ_{out} from the NFM should be gradient as follows [14]:

$$\frac{\partial \phi_{out}}{\partial x} = k_0 \sin \theta_0 \cos \varphi_0 \quad \text{and} \quad \frac{\partial \phi_{out}}{\partial y} = k_0 \sin \theta_0 \sin \varphi_0. \quad (7)$$

Conventionally, the output phase-gradient $\nabla_t \phi_{out}$ is rigorously governed by the phase of the incident far-field wave $\phi_{ext,FF}$, and the distribution of the unit cell transmitted phases $\phi_{UC,FF}$, as defined by

$$\nabla_t \phi_{out} = \nabla_t (\phi_{ext,FF} - \phi_{UC,FF}). \quad (8)$$

Based on the numerical analysis from Fig. 5, a robust approximation is established within the range of $\pi/2 < nk_0t \leq 2\pi$, where the spatial derivative of the total near-field phase $\nabla_t \phi_{UC,NF}$ aligns with that of the radiative component $\nabla_t \phi_{UC,FF}$. Crucially, this phase-gradient alignment implies that the required unit cell phase in the near field, $\phi_{UC,NF}$, can be effectively replaced by $\phi_{UC,FF}$ calculated under conventional plane-wave incidence, as they differ only by a fixed phase offset across the aperture. Consequently, when the unit cells are integrated in the extreme near-field phase of the source, the output phase gradient can be accurately approximated by

$$\nabla_t \phi_{out} = \nabla_t (\phi_{ext,NF} - \phi_{UC,NF}). \quad (9)$$

Accordingly, the required unit cell phase distribution is synthesized by $\phi_{UC,NF} = \phi_{ext,NF} - \phi_{out} + \phi_c$, where ϕ_c represents the phase constants that remain uniform across the NFM aperture. Since the beam tilting direction is fundamentally

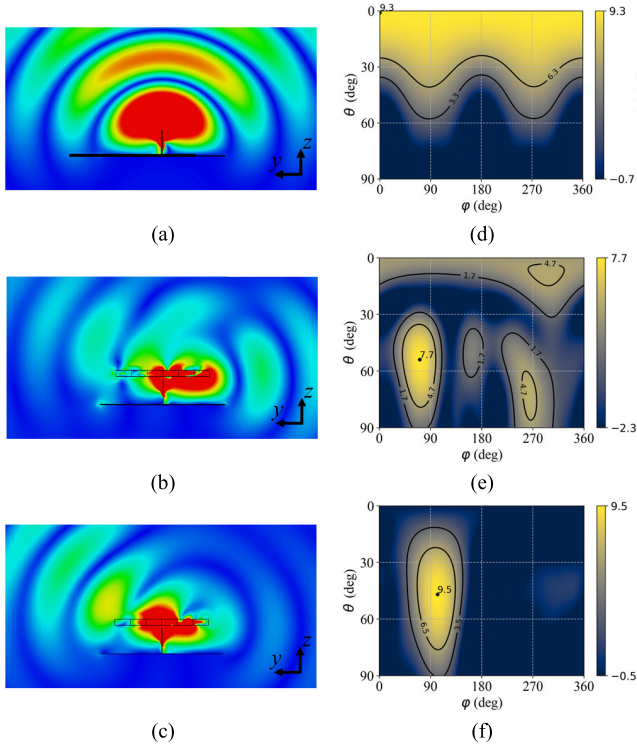


Fig. 7. Numerical investigation of the prephase adjustment on electric field distributions and radiation patterns. (a)–(c) E-field magnitudes for a standalone dipole antenna, loaded NFM with zero prephase, and NFM with $\pi/4$ prephase, respectively. (d)–(f) Corresponding far-field radiation patterns.

determined by the spatial derivative of the phase rather than its absolute value, these constant terms can be neglected during the practical implementation of the required ϕ_{UC} .

Take $\theta_0 = 45^\circ$ and $\varphi_0 = 90^\circ$ as an example, the output phase distribution ϕ_{out} and the transmitted phase ϕ_{UC} of the unit cells are shown in Fig. 6(b) and (c), respectively. The transmitted phase ϕ_{UC} is achieved by adjusting the refractive index n of the unit cells.

As one example, a half-wave dipole resonating at 2.0 GHz is placed a quarter wavelength above a ground plane. The corresponding E-field distribution and radiation patterns are shown in Fig. 7(a) and (d), respectively. The proposed NFM covers the dipole antenna at a distance $d = 7.5$ mm ($0.05\lambda_0$), as the configuration shown in Fig. 1(c), where the period of the unit cells is $p = 25$ mm ($\lambda_0/6$), and the thickness is $t = 10$ mm ($\lambda_0/15$). Since the full 360° phase coverage requires $\Delta n(t/\lambda_0) \geq 1$, the selection of $t = \lambda_0/15$ represents an optimized trade-off: a thinner profile would demand a broader refractive index range Δn , approaching the practical limits of multilayer unit cell realization. This balanced thickness ensures that the NFM remains sufficiently thin while providing the robust phase control required for the targeted 2-D beam tilting.

Based on previous analysis, NFM benefits from its compactness. For a half-wave dipole, an NFM with side length $l = \lambda_0$ interacts with most of the electromagnetic energy. Therefore, the NFM is composed of 6×6 unit cells in total.

However, the NFM readily excites guided waves along the MTS [35], which are scattered and reflected at the truncated edges of the MTS, causing pattern distortion. In the example of beam tilting to $\theta_0 = 45^\circ$ and $\varphi_0 = 90^\circ$, the simulated magnitude of E-fields in the plane $x = 0$ is shown in Fig. 7(b), where

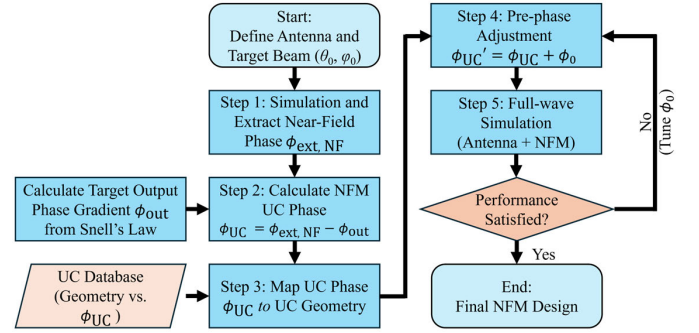


Fig. 8. Systematic design procedure of the proposed compact NFM for 2-D antenna radiation pattern tilting.

scattered fields are found near the edge $y = -\lambda_0/2$. Thereby, the corresponding radiation pattern is highly distorted, as shown in Fig. 7(e). In addition to the intended beam at $\theta_0 = 45^\circ$ and $\varphi_0 = 90^\circ$, two significant spurious beams can be found near $\theta_0 = 0^\circ$, and $\theta_0 = 75^\circ$ and $\varphi_0 = 270^\circ$, caused by the resonance and edge diffraction of the NFM, respectively.

To alleviate the distortion, prephases are introduced. Since the beam tilts by phase gradient, the absolute output phase ϕ_{out} is flexible for shifting any fixed prephase ϕ_0 . By adjusting the absolute output phase of each unit cell while preserving the phase gradient, the propagation constants along the normal and transverse directions are modified. This, in turn, modifies the resonant modes and scattered fields. By optimizing the prephase, unwanted resonances can be suppressed or destroyed, leading to an improved radiation pattern with reduced distortion. In the previous example, a prephase $\pi/4$ is applied uniformly across all unit cells, as shown in Fig. 7(c) and (f). This adjustment results in weaker scattered fields near the edges of the NFM, and the corresponding radiation pattern concentrates the energy toward $\theta_0 = 47^\circ$ and $\varphi_0 = 95^\circ$, closely matching the intended design. Also, the directivity is improved by 1.8 dB. The introduction of a prephase is thus demonstrated to be an effective method for alleviating beam distortion caused by the truncation of a compact NFM while maintaining the desired beam tilting.

To facilitate the transition from theoretical derivation to practical engineering implementation, the comprehensive design methodology for the proposed compact NFM is summarized in the flowchart shown in Fig. 8. The procedure begins by defining the specifications of the source antenna and the target beam directions θ_0 and φ_0 . In Step 1, the near-field phase distribution $\phi_{ext, NF}$ is extracted from the simulation results of the driven antenna. Step 2 involves calculating the required output phase-gradient ϕ_{out} based on the generalized Snell's law to determine the necessary transmission phase ϕ_{UC} for each unit cell. Subsequently, Step 3 maps these required phase values to specific geometric parameters by querying a preestablished unit cell database.

A key innovation in this design flow is Step 4, where prephase adjustments are introduced by adding a phase offset ϕ_0 to the calculated unit cell phases. This adjustment is specifically designed to suppress guided-wave resonances and mitigate edge diffraction at the truncated boundaries of the MTS, which otherwise lead to radiation pattern distortion. Finally, Step 5 involves a full-wave simulation of the integrated antenna-NFM system to verify that the tilting angle and radiation characteristics satisfy the performance requirements before finalizing the design.

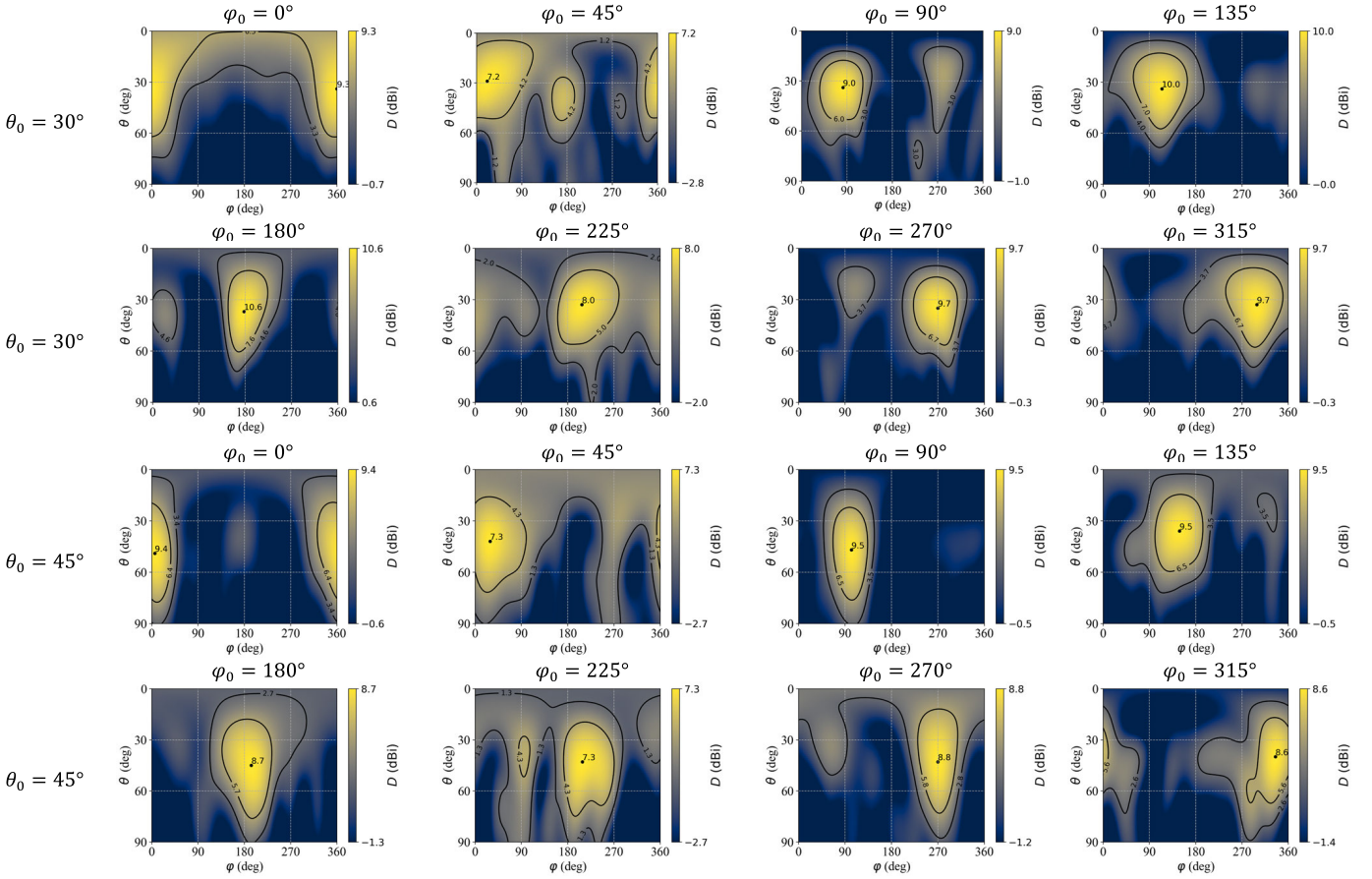


Fig. 9. Radiation patterns of a dipole antenna with NFM composed of ideal materials. Beams directed toward $\theta_0 = 30^\circ$ or 45° and φ_0 steering in 360° with step 45° .

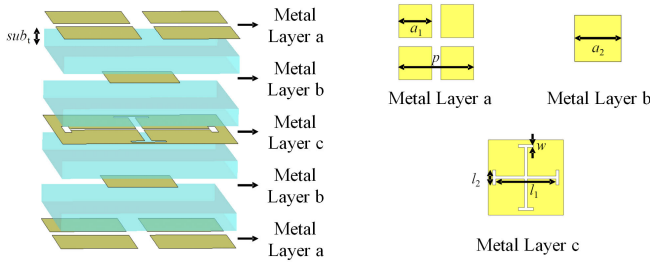


Fig. 10. Schematic of unit cells ($p = 25$ and $sub_t = 1.25$; unit: mm).

Simulated 2-D beam tilting results are presented in Fig. 9, where the beam is designed to point toward $\theta_0 = 30^\circ$ and $\theta_0 = 45^\circ$ while φ_0 is steered in 45° increments across 360° . Minor deviations in the beam directions are attributed to near-field approximations, reflections between the NFM and ground, resonances, and scattered fields from truncated edges of the NFM.

V. NFM DESIGN AND MEASUREMENT

To achieve the desired transmitted phase, a five-metal-layer unit cell is designed as illustrated in Fig. 10. The dielectric substrate is TP615 ($\epsilon_r = 6.15$) with a thickness of 2.5 mm. By sweeping the geometric parameters of the unit cell, a set of extracted effective permittivity ϵ_r , permeability μ_r , refraction index n , and normalized impedance z covering the

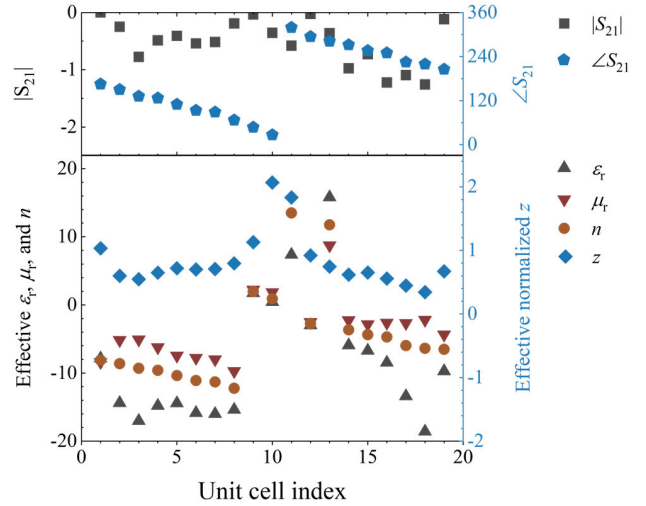


Fig. 11. Transmitted magnitude and phase of unit cells at 2.0 GHz and the corresponding extracted effective permittivity ϵ_r , permeability μ_r , refractive index n , and normalized impedance z .

full phase range with high transmission at 2 GHz is obtained, as summarized in Table 1 and Fig. 11 [36]. For each unit cell, the ϵ_r and μ_r are all close to each other, maintaining a near-unity normalized impedance to minimize reflection. Moreover, the absolute values of ϵ_r and μ_r vary for achieving a full-range phase shift. Although most refractive indices are negative and

TABLE I
GEOMETRY PARAMETERS OF THE UNIT CELLS (UNIT:MM)

Index	w	a ₁	a ₂	l ₁	l ₂
1	1	8.4	15	18	2.0
2	1	8.2	15	18	2.0
3	1	8	15	17	2.0
4	1	8	15	16	2.0
5	1	7.8	15	15	2.0
6	1	7.5	15	15	2
7	1	7.4	15	15	2
8	1	7.2	15	15	2
9	1	7.5	14	15	2
10	1	8	10	16	4
11	1	9.5	15	18.5	16
12	1	9.5	15	18	15.5
13	2	9.5	15	18.5	16.2
14	1	9.5	15	18	15
15	1	9.2	15	18	15
16	1	9.2	15	18	14.5
17	1	9.2	15	18	12
18	1	9.2	15	18	10
19	1	9	15	18	12

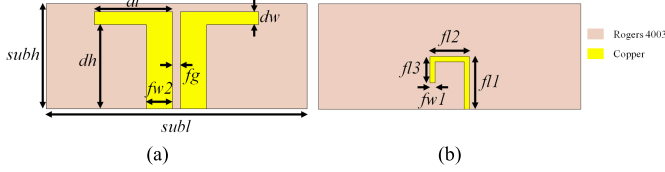


Fig. 12. Dipole antenna. (a) Front view. (b) Back view.



Fig. 13. Fabricated dipole antenna integrated with the proposed NFM. (a) Side view. (b) Top view.

TABLE II
GEOMETRY PARAMETERS OF THE DIPOLE ANTENNA (UNIT:MM)

dl	30	dw	5	dh	32
fg	3	subl	100	subh	40
fw1	2	fw2	10	fl1	20
fl2	15	fl3	10		

exhibit abrupt changes at certain points, the transmitted phase can be wrapped and change gradually in the range of 0° – 360° range. The unit cells are well prepared for the following beam tilting NFM design.

A half-wave dipole is designed, resonating at 2 GHz above a ground, as shown in Fig. 12. The dielectric substrate is Rogers 4003 ($\epsilon_r = 3.55$) with a thickness of 1.5 mm. The remaining parameters are summarized in Table II. The fabricated NFM, composed of the previously designed unit cells and satisfying the required unit cell transmitted phase distribution for desired beam tilting, is mounted with $d = 0.05\lambda_0$ above the dipole

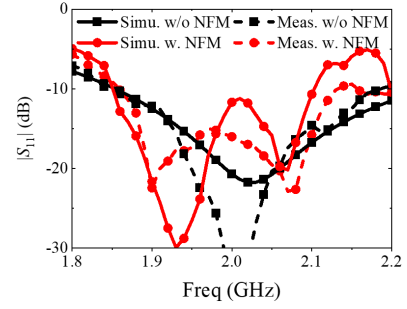


Fig. 14. Simulated and measured $|S_{11}|$ with and without the NFM.

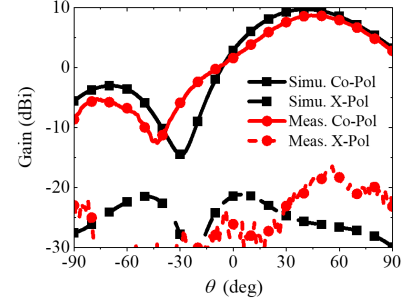


Fig. 15. Simulated and measured radiation patterns of the dipole antenna with the NFM in the $\varphi = 90^\circ$ plane.

TABLE III
COMPARISON WITH COMPACT NEAR-FIELD MTS DESIGNS

	Distance $d (\lambda_0)$	Area (λ_0^2)	Far Field Approximation	MTS Function
[15]	0.31	$3.0^2\pi$	Y ¹	Gain Enhancement
[16]	0.26	$3.0^2\pi$	Y ¹	Gain Enhancement
[17]	0.29	6.0×6.0	Y ¹	2-D Steering
[18]	0.30	6.0×6.0	Y ¹	Gain Enhancement
[19]	0.44	$>20 \times 20$	Y ¹	1-D Steering
[20]	0.18	0.8×3.9	Y ²	1-D Steering
[21]	1	$3.5^2\pi$	Y ²	2-D Steering
[22]	0.4	$3.3^2\pi$	Y ²	2-D Steering
[23]	1.2	2.4×2.4	Y ²	1-D Steering
Proposed	0.05	1.0×1.0	N	2-D Steering

- 1: By applying partially reflection surface to form an equal phase plane.
2: By spherical wave approximation

antenna. The overall size of the NFM is 150×150 mm ($\lambda_0 \times \lambda_0$ at 2 GHz), consisting of 6×6 unit cells. A prototype is fabricated with the goal of tilting the beam toward $\theta_0 = 45^\circ$ and $\varphi_0 = 90^\circ$, as shown in Fig. 13.

Fig. 14 shows the simulated and measured $|S_{11}|$. By tuning the prephase, the dipole antenna can remain well-matched when mounted with the NFM at 2 GHz, without any modification to the antenna geometry. The discrepancies between measured and simulated results are attributed to air gaps between the multiple layers of the NFM. Furthermore, Fig. 15 shows the simulated and measured radiation patterns, with a measured realized gain of 8.7 dBi at $\theta_0 = 44^\circ$. Overall, the agreement between measured and simulated results validates the proposed approach to design compact NFM for beam tilting.

TABLE IV
COMPARISON WITH NEAR-FIELD METAMATERIAL LOADINGS

	Thickness (λ_0)	E-plane Tilting	H-plane Tilting
[8]	0.54	[6°, 19°]	N
[9]	0.56	[-25°, 25°]	N
[10]	0.84	[8°, 31°]	N
[11]	0.15	N	[-20°, -35°]
Proposed	0.07	θ in [-45°, 45°], ϕ in [0°, 360°]	

The proposed NFM, positioned in the extreme near fields of the antenna, achieves the 2-D beam tilting by accounting for the near-field components during the design. The performance is summarized and compared with other near-field and compact MTS designs in Table III. Conventionally, MTS placed in the near fields of an antenna involves near-field components and a nonuniform phase distribution of the incident wave. Therefore, most prior works place the MTS at least a quarter wavelength away from the antenna to ensure that the incident wave is dominated by radiated components, thereby avoiding the complexities of near-field interactions. Then, a partially reflecting surface is employed before the beam tilting MTSs to produce uniform incident phases.

Another approach approximates the antenna as a point source and regards the incident wave of MTSs as spherical waves. In this proposed method, with the study of near-field components of an antenna and the demonstration of applying phase gradient for beam tilting and prephases for alleviating near-field effects, the proposed design achieves 2-D beam tilting and benefits from reduced distance d between the MTS and the antenna, as well as the compact dimensions of the MTS.

Furthermore, the proposed NFM is compared with another beam tilting method by nearly placed metamaterial loadings in Table IV. The essential principle of the loadings is tuning the output propagation direction from the loadings. Typically, a thick metamaterial block is required to achieve sufficient phase delay. In addition, with the far-field approximation, the method is qualitative and requires trial and error for desired beam tilting.

VI. CONCLUSION

The near fields of a dipole antenna have been studied for providing critical insights into the near-field characteristics, enabling the effective design of compact NFMs for beam control. The approximately identical slopes of the transmitted near- and far-field phases for the parallel-polarized component have been shown to guarantee beam tilting through phase gradients imposed on the NFM. The feasibility of compact NFMs has been demonstrated through ideal materials for 2-D beam tilting. Simulation and measurement results validated the prototype's ability to achieve the desired beam tilting at $\theta_0 = 44^\circ$ and $\phi_0 = 90^\circ$ with $\lambda_0 \times \lambda_0$ aperture using PCB-based unit cells mounted $0.05\lambda_0$ above a dipole antenna. The results demonstrate a promising approach for advanced radiation pattern control using a low-profile and compact aperture, potentially enabling the integration of directionally flexible antennas into next-generation communication systems.

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Bo Zhang (Member, IEEE) received the B.Eng. degree in electromagnetic field and wireless technology from the University of Electronic Science and Technology of China, Chengdu, China, in 2019, and the M.S. and Ph.D. degrees from the National University of Singapore, Singapore, in 2020 and 2025, respectively.

Since 2025, he has been a Research Fellow with the Department of Electrical and Computer Engineering, National University of Singapore. His current research interests include metamaterials,

metasurfaces, antenna radiation patterns, near fields, multiple-input–multiple-output (MIMO) antennas, and machine learning applications in electromagnetics.

Dr. Zhang was a recipient of the Honorable Mention of the 2021 IEEE Antennas and Propagation Society (AP-S) Student Paper Competition and the Best Student Paper Award from the 2025 IEEE Asia-Pacific Conference on Antennas and Propagation (APCAP). He serves as a reviewer for IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION.



Zhi Ning Chen (Fellow, IEEE) received the B.Eng., M.Eng., and Ph.D. degrees in electrical engineering from the Institute of Communications Engineering (ICE), Nanjing, China, in 1985, 1988, and 1993, respectively, and the second Ph.D. degree from the University of Tsukuba, Tsukuba, Japan, in 2003.

From 1988 to 1997, he worked with ICE, Southeast University, Nanjing, China, and the City University of Hong Kong, Hong Kong. He conducted his research with the University of Tsukuba from 1997 to 1999 and visited the University in

2001 and 2004, all sponsored by the Japan Society for the Promotion of Science Fellowship Programs. In 2004, he worked at IBM Watson Research Center, Yorktown, NY, USA, as an Academic Visitor. In 2013, he visited the "Laboratoire des Signaux et Systèmes," UMR8506 CNRS-Supelec-University Paris Sud, Gif-sur-Yvette, France, as a Senior DIGITEO Guest Scientist. In 2015, he visited the Center for Northeast Asian Studies of Tohoku University, Sendai, Japan, as a Senior Visiting Professor. From 1999 to 2016, he worked at the Institute for Infocomm Research, Singapore, in various capacities, including as a Principal Scientist, the Department Head, the Manager, and a Technical Advisor. In 2012, he joined the Department of Electrical and Computer Engineering, National University of Singapore, as a Tenured Provost's Chair Professor. He was appointed as an Industry Program Director and the Director of the Advanced Research and Technology Innovation Center. He has/held concurrent guest professorships at 15 + local and overseas universities. In addition, he has provided consultancy services to local and overseas companies as a Technical Advisor, a Guest Professor, and the Chief Scientist. He has published 720 papers and six books entitled *Broadband Planar Antennas* (Wiley 2005), *UWB Wireless Communication* (Wiley 2006), *Antennas for Portable Devices* (Wiley 2007), *Antennas for Base Stations in Wireless Communications* (McGraw-Hill 2009), *Handbook of Antenna Technologies* with 76 chapters (Springer References 2016 as the Editor-in-Chief), and *Substrate-Integrated Millimeter-Wave Antennas for Next-generation Communications and Radars* (Wiley and IEEE Press 2021) plus chapters to other books. He holds 36 granted and filed patents and has completed 43 licensing deals with the industry. His current interests include the translational research of electromagnetic metamaterials and the application of prior knowledge-guided machine learning to antenna engineering.

Dr. Chen was elevated to a fellow of the Academy of Engineering, Singapore, in 2019, and his contributions to small and broadband antennas for wireless applications in 2007. Currently, he serves on the IEEE Future Networks Initiative Technology Focus Editorial Board as a member. He was a recipient of the EurAAP Antenna Award in 2025, the IEEE Antennas and Propagation Society John Kraus Antenna Award in 2021, the ASEAN Outstanding Engineering Achievement Award in 2013, and the Institution of Engineers Singapore Prestigious Engineering Achievement Awards in 2006, 2013 (two awards), and 2014, and 35 + best paper awards with his students. He served as the Chair of the IEEE Singapore MTT/AP Chapter in 2008 and as the Founding Chair of the IEEE Singapore RFID Chapter from 2016 to 2017. He is the Founding General Chair of the International Workshop on Antenna Technology in 2005, the International Symposium on InfoComm and Mechatronics Technology in Biomedical and Healthcare Application in 2010, the International Microwave Forum in 2010, the Asia-Pacific Conference on Antennas and Propagation in 2012, and the Marina Forum in 2020. He has been involved in international events as general chairs and chairs for technical program committees and advisory committees. He is the General Chair of the 2021 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting. He has delivered 180 + keynotes, plenaries, and invited speeches at international academic and industry events. He served the IEEE Council on RFID as a Distinguished Lecturer and as a Founding Vice President from 2015 to 2020. He serves on IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION as an Associate Editor and has served the IEEE Antennas and Propagation Society as a Distinguished Lecturer, as well as a member of the AdCom Committee, the Fellow Committee, and the New Directions Committee.